

# NIGEL VAN DER HORST

+65 97802510 | [nigel.vdh@u.nus.edu](mailto:nigel.vdh@u.nus.edu) | Singapore Citizen  
[www.linkedin.com/in/nigel-vanderhorst](https://www.linkedin.com/in/nigel-vanderhorst) | [vdhorstnigel.github.io](https://vdhorstnigel.github.io)

I am a driven NUS Electrical Engineering undergraduate with a passion for embedded systems, computer networking, and cybersecurity. I have hands-on experience across diverse internships - automating data integration into SCADA at Shopee, building an MVP risk-analysis platform at Cysecure, and deploying call-centre networks for Terra Systems – which have sharpened my problem-solving and operational skills. Proactive and collaborative, I excel at turning technical challenges into practical solutions.

## EDUCATION

---

|                                                                                                                                        |                            |
|----------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| <b>National University of Singapore</b><br><b>Bachelor of Engineering (Honours) - Electrical Engineering with a Minor in Computing</b> | <b>Aug 2022 – Dec 2025</b> |
|----------------------------------------------------------------------------------------------------------------------------------------|----------------------------|

- GPA: 4.22/5.0
- Relevant coursework: Introduction to Programming methodology (python), Microcontroller Programming and Interfacing, Computer networks, Datascience for IoT, Hardware for IoT

|                                                                        |                            |
|------------------------------------------------------------------------|----------------------------|
| <b>Singapore Polytechnic</b><br><b>Diploma in Computer Engineering</b> | <b>Apr 2017 - May 2020</b> |
|------------------------------------------------------------------------|----------------------------|

- GPA: 3.64/4.0 (Director's Honour Roll 2019/2020)
- Relevant coursework: Computing Networking, Server Management, Network Security, Cyber Security

## WORK EXPERIENCE

---

|                                             |                            |
|---------------------------------------------|----------------------------|
| <b>Shopee   Operation Automation Intern</b> | <b>Jan 2025 - Apr 2025</b> |
|---------------------------------------------|----------------------------|

- Designed and deployed Netdata for network monitoring across 4 sorting centres, **reducing investigation downtime by 30%**
- Leveraged AWS Lambda and DynamoDB to automate 7 critical alert integrations into native Seataalk app, **reducing manual intervention and incident response time by 20%**
- Integrated network metrics into in-house SCADA systems, enabling correlation of OT/IT issues

|                                                   |                            |
|---------------------------------------------------|----------------------------|
| <b>Cysecure Pte Ltd   Systems Engineer Intern</b> | <b>Jun 2024 - Aug 2024</b> |
|---------------------------------------------------|----------------------------|

- Spearheaded development of an MVP risk analysis platform for non-technical personnel, enabling cost analysis for risks identified

|                                                   |                            |
|---------------------------------------------------|----------------------------|
| <b>Terra Systems   Network Engineering Intern</b> | <b>May 2019 - Oct 2019</b> |
|---------------------------------------------------|----------------------------|

- Troubleshoot network issues for 10+ clients, **maintaining 99.99% SLA compliance**
- Set up 2 call centres using Cisco UCM/UCCX and hardened security via firewall policies

## LEADERSHIP ROLES

---

|                                                                           |                            |
|---------------------------------------------------------------------------|----------------------------|
| <b>NUS College of Design and Engineering (CDE) Club, Content Director</b> | <b>Sep 2022 - Sep 2023</b> |
|---------------------------------------------------------------------------|----------------------------|

- Oversaw building and bolstering CDE brand identity as a new faculty of NUS for over 2,000 students
- Led a team of 5 to create publicity, growing social media platforms leading to a 65% boost in followers

## ACADEMIC PROJECTS

---

|                                                                         |                           |
|-------------------------------------------------------------------------|---------------------------|
| <b>Capstone Project, Machine Learning &amp; Embedded hardware ESP32</b> | <b>Jan 2025- Dec 2025</b> |
|-------------------------------------------------------------------------|---------------------------|

- Developed an embedded electronic nose to detect food spoilage
- Integrated edge impulse model and modified K-means Cluster algorithm for anomaly detection

## ADDITIONAL INFORMATION

- 
- **Languages:** English – Native proficiency, Chinese - intermediate working proficiency
  - **Programming:** Python, Assembly, C
  - **Networking:** Cisco Routing & Switching, Network security, VoIP (UCM/UCCX)
  - **System admin:** MySQL, Window Servers, Linux (Ubuntu)
  - **Cloud/DevOps:** AWS Lambda, DynamoDB, Linux, Netdata